Higher Nationcccal Diploma in Civil Engineering

Second Year, Second Semester Examination - 2022

CE 2212-Structural Design

Q1)

A four-span continuous one way spanning solid slab floor system is shown in Fig Q1[15]. The following data are given regarding the system[16]:

Load/Strength	**Value**
Characteristic dead load (excluding self-weight)	12.0 KN/m^2
Characteristic imposed load	5.0KN/m^2
Characteristic Strength of concrete f_{cu}	40N/mm^2
Characteristic strength of reinforcing steel f_y	460N/mm^2
Cover to the steel	40mm

a. Find a suitable slab thickness using (span /effective depth) ratio as 20. (2 marks)

b. Show the validity of using the relevant table from BS 8110, for the calculation of bending moment and shear force for the above solid slab system. (3 marks)

c. Determine the maximum sagging moment of the slab. (7 marks)

d. Calculate the main reinforcement for the maximum sagging moment. (4 marks)

e. Check for deflection. (4 marks)

(Total 20 marks)

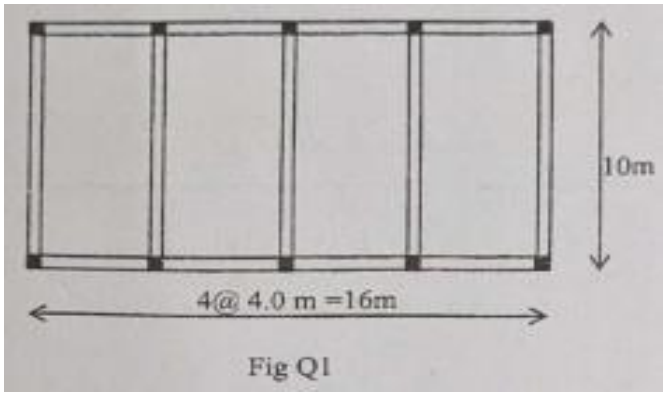


Fig Q1

Q2)

a. State the classification of columns and define them36. (4 marks)

b. Classify the 400mm x 250mm column shown in fig Q2 if the structure is braced. (5 marks)

c. The column shown in fig Q2 carries an axial load of 1200kN and a moment of 60kNm about the x-x axis and 40kNm about the y-y axis.Design the longitudinal and shear reinforcement assuming characteristic strength of concrete and steel as 30 N/mm², 460N/mm² and 250N/mm². The cover to all steel is 30mm. (11 marks)

(Total 20 marks)

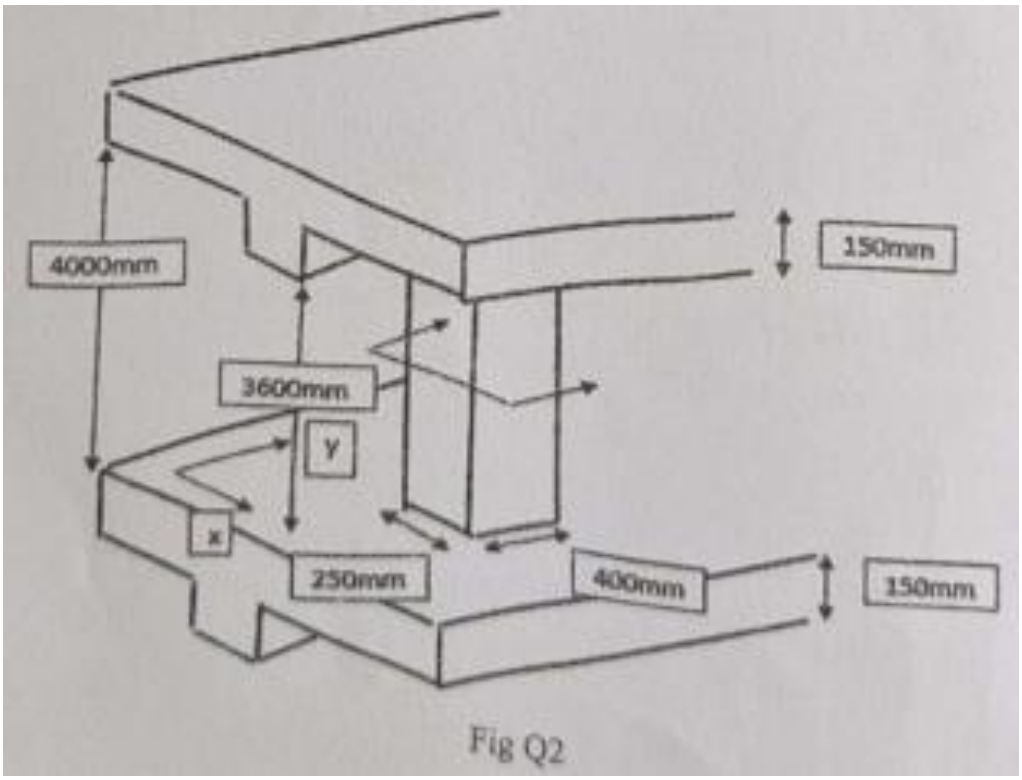


Fig Q2

Q3)

Design the bending and shear reinforcement for a simply supported reinforced concrete beam having a span, breadth and effective depth of 8.5m, 250mm and 450mm respectively.The characteristic values of uniformly distributed dead and imposed loads are $8kNm^{-1}$ and $14kNm^{-1}$ respectively.The characteristic strengths of the concrete and steel may be taken as $40N/mm^2$ and $460N/mm^2$.Assume the cover to the main reinforcement bar as 40mm and the diameter of main reinforcement bar as 20mm.

(Total 20 marks)

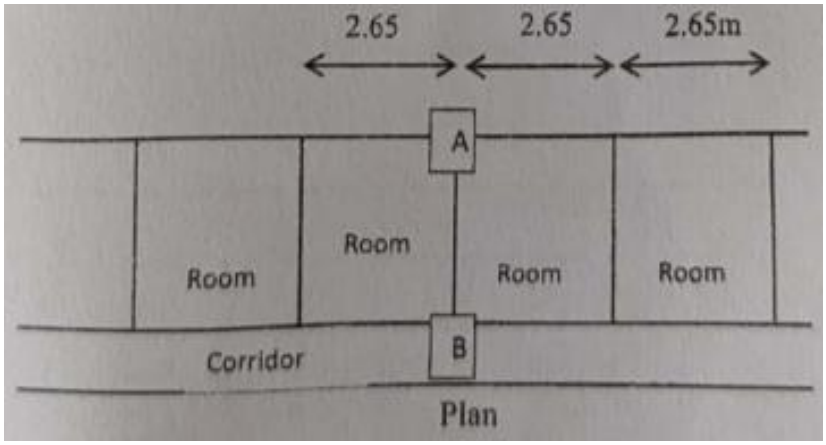
Q4)

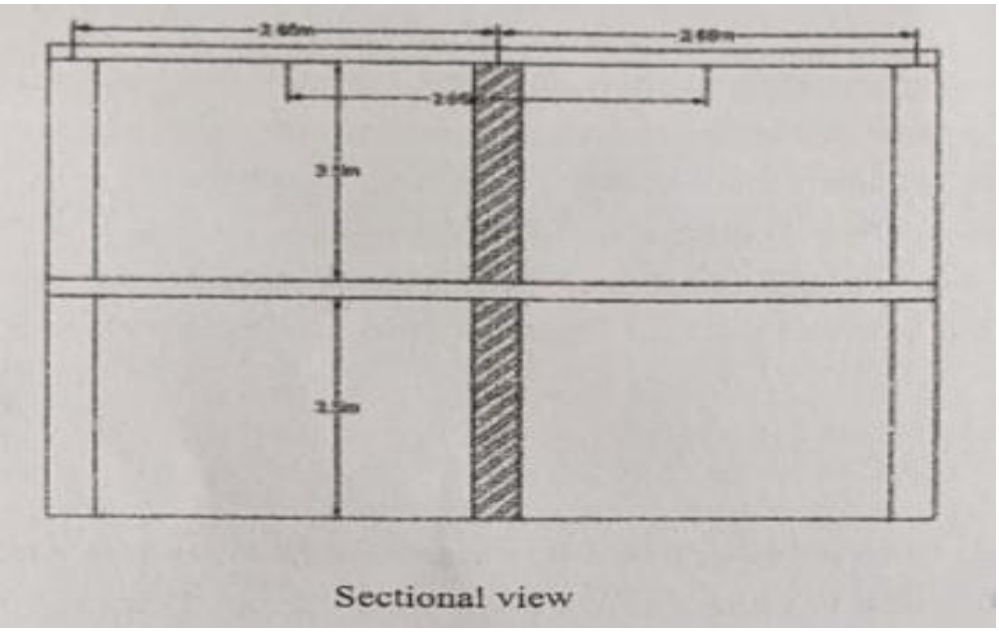
Design a **lap joint** between two plates with the grade of **S275** , each of width **120mm** if the thickness of one plate is **16mm** and another plate is **12mm**. This joint has to transfer a design load of **160kN**.

(Total 20 marks)

Q5)

A two story hotel building has a **100mm thick cast in-situ one way spanning slab** for the roof and the 1st floor. Floor to floor clear height of each story has **3.5m**. Consider that total dead load of each slab is **$3.0kN/m^2$** and imposed loads of roof and floor are **$1.5kN/m^2$ and $2kN/m^2$** respectively. Design the typical internal brick wall (AB) at the ground floor level (ignore lateral load effect) considering the similar floor plan at ground floor and 1st floor. Assume that the effective height of the wall is given by **0.75x clear height of the wall**.





Q6)

a. List down four characteristics which influence the timber design71. (4 marks) 72

b. A timber column 200mm x 200mm is required to carry a load of 210kN. The load has been transferred to the column by timber joists such that the end restraint conditions top and bottom may be taken as restrained in position but not in direction. The height of column is 2.8m and the timber may be taken as strength class . The load may be considered as short term duration at service class 3. (16 marks)

(Total 20 marks)

Table 3.13 Values of A_w/s_v

Diameter (mm)	Spacing of links (mm)										
	85	90	100	125	150	175	200	225	250	275	300
8	1.183	1.118	1.006	0.805	0.671	0.575	0.503	0.447	0.402	0.366	0.335
10	1.847	1.744	1.57	1.256	1.047	0.897	0.785	0.698	0.628	0.571	0.523
12	2.659	2.511	2.26	1.808	1.507	1.291	1.13	1.004	0.904	0.822	0.753
16	4.729	4.467	4.02	3.216	2.68	2.297	2.01	1.787	1.608	1.462	1.34

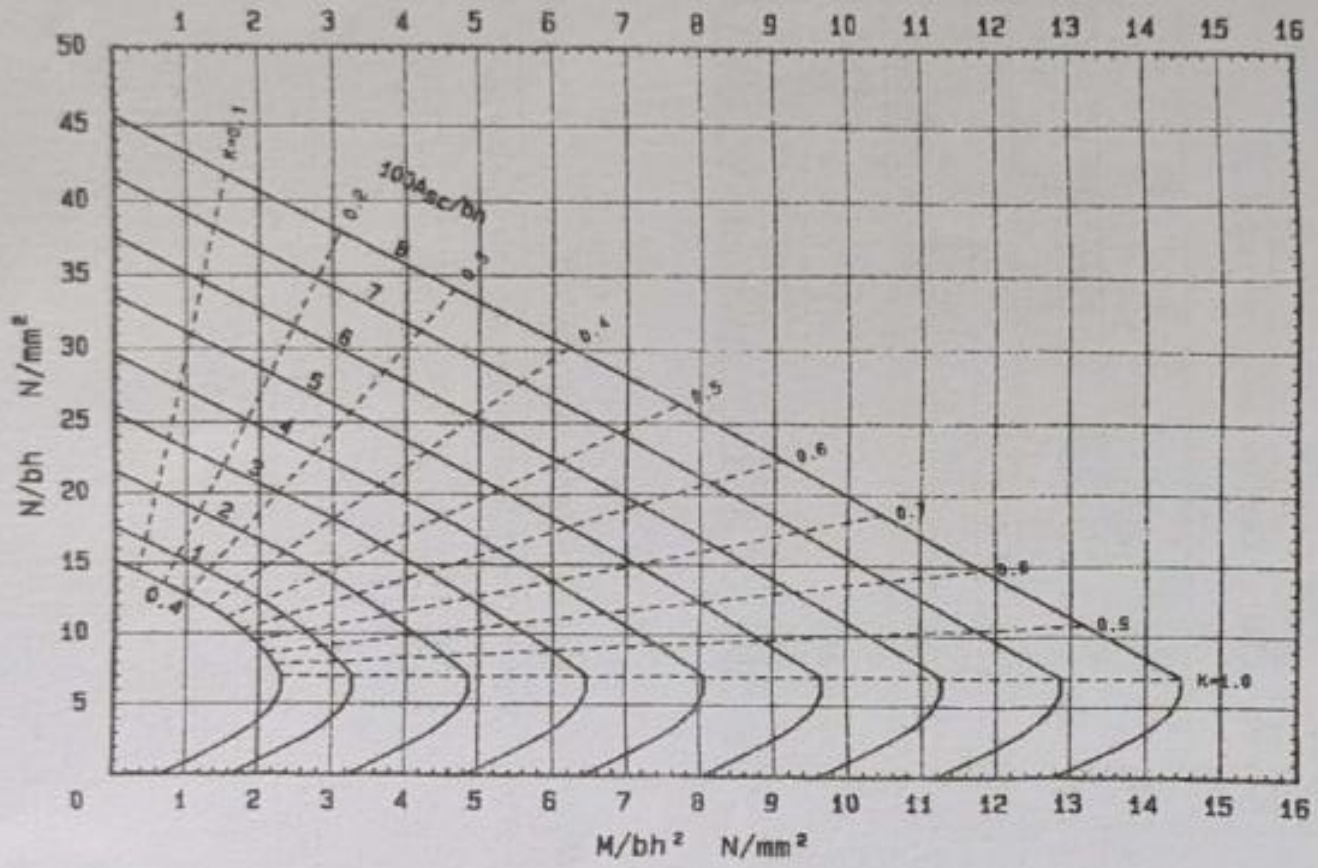
Area's of Reinforcement

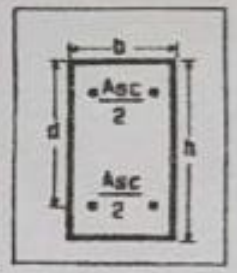
Cross-sectional areas of groups of bars (mm^2)

Bar Size	Number of Bars * Denotes non-preferred sizes									
	1	2	3	4	5	6	7	8	9	10
6mm	28.3	56.5	84.8	113	141	170	198	226	255	283
8mm	50.3	100	151	201	251	302	352	402	452	503
10mm	78.5	157	236	314	339	471	550	628	707	785
12mm	113.1	226	339	452	566	679	792	905	1718	1131
16mm	201.1	402	603	804	1005	1206	1407	1608	1810	2011
20mm	314.2	628	942	1257	1571	1885	2199	2513	2827	3142
25mm	490.9	982	1473	1963	2454	2945	3436	3927	4418	4909
32mm	804.2	1608	2413	3217	4021	4825	5630	6434	7238	8042
40mm	1257.0	2513	3770	5026	6283	7540	8790	10053	11310	12566
50mm*	1963.5	3927	5890	7854	9812	11781	13744	15708	17671	19635

Cross-sectional areas for Specific Spacings in mm^2 / Metre Width

Bar Size	Spacing of Bars (Millimetres) * Denotes non-preferred sizes									
	75	100	125	150	175	200	225	250	275	300
6mm *	377	283	226	188	162	141	125	113	103	94
8mm	670	503	402	335	287	251	223	201	183	168
10mm	1047	786	628	524	449	393	348	314	285	262
12mm	1508	1131	905	754	646	565	503	452	411	377
16mm	2681	2011	1608	1340	1149	1005	894	801	731	670
20mm	4189	3142	2513	2094	1795	1571	1396	1257	1143	1047
25mm	6545	4909	3927	3272	2805	2454	2182	1963	1785	1636
32mm	-	8042	6434	5362	4596	4021	3574	3217	2924	2681
40mm	-	-	10053	8378	7181	6283	5587	5027	4571	4189
50mm*	-	-	-	13090	11220	9617	8727	7854	7140	6545





f_{cu}	30
f_y	460
d/h	0.90

Rectangular columns

Chart No. 29

Table 2 — Characteristic compressive strength of masonry, f_k, in N/mm²

a) Constructed with standard format bricks

Mortar designation	Compressive strength of unit (N/mm²)								
	5	10	15	20	27.5	35	50	70	100
(i)	2.5	4.4	6.0	7.4	9.2	11.4	15.0	19.2	24.0
(ii)	2.5	4.2	5.3	6.4	7.9	9.4	12.2	15.1	18.2
(iii)	2.5	4.1	5.0	5.8	7.1	8.5	10.6	13.1	15.5
(iv)	2.2	3.5	4.4	5.2	6.2	7.3	9.0	10.8	12.7

Table 4a) — Partial safety factors for material strength — Compression, γ_m

		Category of construction control	
		Special	Normal
Category of manufacturing control of structural units	Special	2.5	3.1
	Normal	2.8	3.5

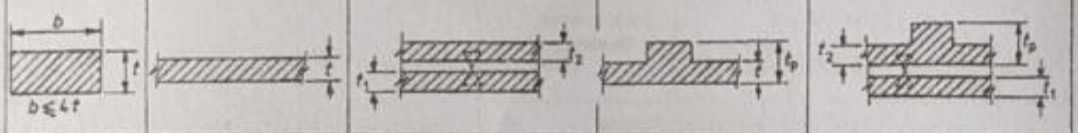
Column	Single-leaf wall	Cavity wall	Walls stiffened by piers	
			Single-leaf	Cavity
Plan shapes				
				
Effective thickness				
t or b, depending on direction of bending	t	the greatest of: a) $2/3\ (t_1 + t_2)$; or b) t_1; or c) t_2	$t \times k$ where K is the stiffness coefficient from Table 5	the greatest of: a) $2/3\ (t_1 + Kt_2)$; or b) t_1; or c) Kt_2

Figure 3 — Effective thickness of columns and walls

Table 7 — Capacity reduction factor, β

Slenderness ratio h_{ef}/t_{ef}	Eccentricity at top of wall, e_t			
	Up to $0.05t$ (see note 1)	$0.1t$	$0.2t$	$0.3t$
0	1.00	0.88	0.66	0.44
6	1.00	0.88	0.66	0.44
8	1.00	0.88	0.66	0.44
10	0.97	0.88	0.66	0.44
12	0.93	0.87	0.66	0.44
14	0.89	0.83	0.66	0.44
16	0.83	0.77	0.64	0.44
18	0.77	0.70	0.57	0.44
20	0.70	0.64	0.51	0.37
22	0.62	0.56	0.43	0.30
24	0.53	0.47	0.34	
26	0.45	0.38		
27	0.40	0.33		

NOTE 1 It is not necessary to consider the effects of eccentricities up to and including $0.05t$.

NOTE 2 Linear interpolation between eccentricities and slenderness ratios is permitted.

NOTE 3 The derivation of β is given in Appendix B.

32.2.1 ***Design vertical load resistance of walls***

The design vertical load resistance of a wall per unit length is given by:

$$\frac{\beta t f_k}{\gamma_m}$$

where

β is a capacity reduction factor allowing for the effects of slenderness and eccentricity and is obtained from Table 7;

f_k is the characteristic strength of the masonry obtained from clause **23**;

γ_m is the appropriate partial safety factor for the material obtained from clause **27** Table 4a);

t is the thickness of the wall.

p_{bs} is the bearing strength of the connected ply
= 460N/mm² for S275
= 550N/mm² for S355

Table 32. Strength of bolts in clearance holes

	Bolt grade		Other grades of fasteners
	gr 4.6	gr 8.8	
	N/mm²	N/mm²	N/mm²
Shear strength, p_s	160	375	$0.48U_f$ but $< 0.69\ Y_f$
Bearing strength, p_{bb} (but see table 33)	460	1035	$0.72\ (U_f + Y_f)$
Tension strength, p_t	195	450	$0.58U_f$ but $< 0.83\ Y_f$

Y_f is the specified minimum yield strength of the fastener.

U_f is the specified minimum ultimate tensile strength of the fastener.

Table 33. Bearing strength of connected parts for ordinary bolts in clearance holes, p_{bs}

Steel to BS 4360			Other grades of steel
gr 43	gr 50	gr 55	
N/mm²	N/mm²	N/mm²	N/mm²
460	550	650	$0.65(U_s + Y_s)$

Y_s is the specified minimum yield strength.

U_s is the specified minimum ultimate tensile strength.